

Lab Experiment 3

Subject - Machine Learning Theory and Practice

Subject Code - E1CAB 613

1. Dataset

Concrete Compressive Strength

<https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength>

2. Objective

Understanding Simple and Multiple Linear regression.

3. Lab Question

Perform the following tasks using Python.

Tasks & Marks Distribution (Total: 10 Marks)

Task 1: Exploratory Data Analysis (EDA) (2 Marks)

- Loading the dataset
- Displaying important information about the dataset such as dimensions, summary statistics etc.
- Visualization of the data

Task 2: Data Cleaning & Transformation (2 Marks)

- Outlier detection
- Handling missing values
- Feature Scaling

Task 3: Simple Linear Regression (2 Marks)

- Select one independent variable and build a Simple Linear Regression model using sklearn.
- Split the dataset into training and testing sets.
- Plot the regression line.
- Evaluate the model using R^2 score, MAE, MSE and RMSE.

Task 4: Multiple Linear Regression (2 Marks)

- Build a Multiple Linear Regression model using multiple features.
- Predict the target variable on the test set.
- Plot the regression line.
- Evaluate the model using R^2 score, MAE, MSE and RMSE.

Marks Distribution Summary

Task	Description	Marks
1	Exploratory Data Analysis (EDA)	2
2	Data Cleaning & Transformation	2
3	Simple Linear Regression	2
4	Multiple Linear Regression	2
	Timely submission	2
	Total	10

General Instructions

1. Ensure that your code includes relevant comments to enhance readability and understanding. Subsequently, upload your code to GitHub for version control and collaborative access.
2. Include descriptive comments within the code, explaining its functionality and logic.
3. Attach a PDF document named "your_register_number_Lab_No.pdf" to the submission. The PDF document should include screenshots of the code and the output screen.
4. Kindly attach the Github link to your lab exercise as well.